Spectrometer – Dispersive Power of Prism

Aim

To determine the refractive indices of various colours.
To calculate the dispersive power of a prism.

Apparatus

Spectrometer, prism, prism clamp, mercury vapour lamp, lens.

Principle

When a beam of light strikes the surface of transparent material (glass, water, quartz crystal, etc.), a portion of the light is transmitted and another portion is reflected. The transmitted light ray has a small deviation of the path from the incident angle. This is called refraction.

Refraction is due to the change in the speed of light while passing through the medium. It is given by Snell's Law.

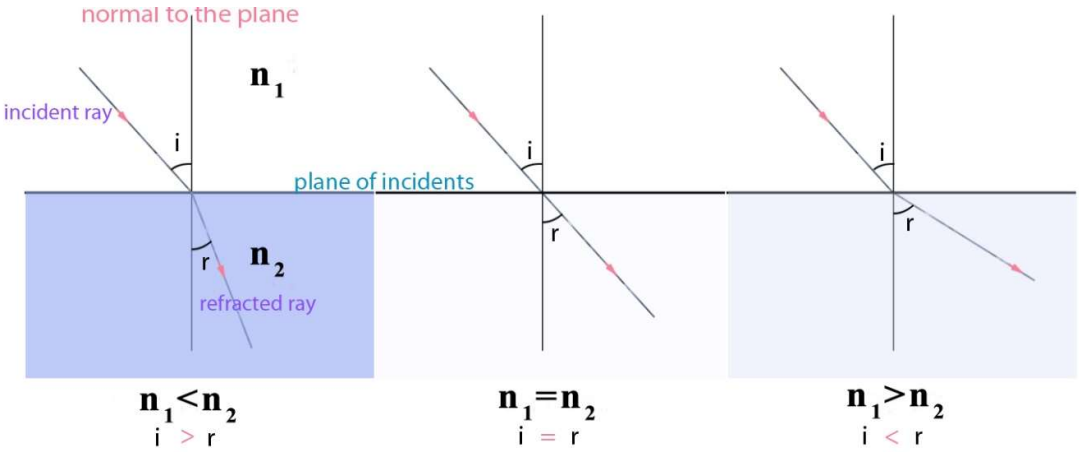
$$\frac{\sin(i)}{\sin(r)} = \frac{n_2}{n_1} \qquad (1)$$

Where i is the angle of incidence, r is the angle of refraction, n_1 is the refractive index of the first face and n_2 is the refractive index of the second face.

The speed of light on both faces is related to the equation

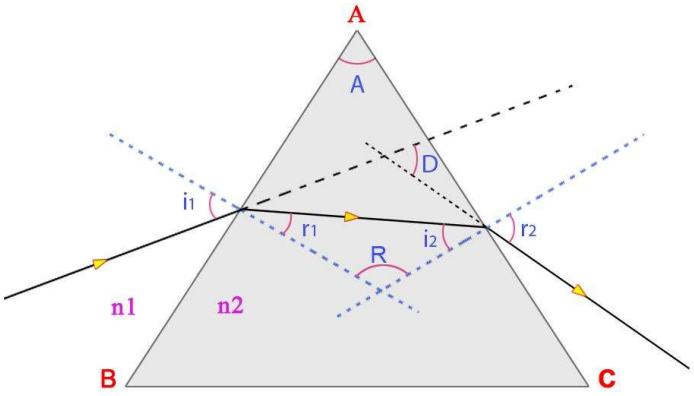
$$\frac{c_1}{c_2} = \frac{n_2}{n_1} \qquad (2)$$

c_1 is the velocity of the wave in the first face and c_2 is the velocity of the wave in the second face.



The above figure illustrates the change in refracted angle with respect to the refractive index.

Consider a prism of angle A and refractive index n_2. Let i_1 and r_1 be the incident and refracted ray from face AB, and i_2 and r_2 be the incident and emerged ray from the second face AC.



Dispersive Power of prism

The refractive index of the material of the prism can be calculated by the equation.

$$n = \frac{\sin\left(\frac{A+D}{2}\right)}{\sin\left(\frac{A}{2}\right)} \qquad (3)$$

Where D is the angle of minimum deviation. Here, D is different for different colours.

Consider two colours – green and violet. The corresponding minimum deviation is D_G and D_V, and the corresponding refractive indices are:

$$n_G = \frac{\sin\left(\frac{A+D_G}{2}\right)}{\sin\left(\frac{A}{2}\right)} \; n_V = \frac{\sin\left(\frac{A+D_V}{2}\right)}{\sin\left(\frac{A}{2}\right)} \qquad (4)$$

Therefore, the dispersive power is

$$\omega = \frac{(n_V - n_G)}{n} \qquad (5)$$

Where,

$$n = \frac{(n_V + n_G)}{2} \qquad (6)$$

Controls

Switches
Switch On/Off Light: Used to switch on/off the light.
Place Prism/Remove Prism: This switch is used to place the prism on the prism table or remove the prism from the prism table.

Slider
Slit focus: This slider is used to focus the slit while looking through the telescope.
Slit width: Using this slider, the width of the slit can be adjusted.
Telescope: Using this slider, one can move the telescope from its position.
Vernier Table: Vernier table can be rotated using this slider.

Fine Angle adjustment
Telescope: This is used to fine adjust the telescope.
Vernier Table: Using this slider, we can rotate a fine angle.

Measurements
Here we get the zoomed view of vernier I and II by placing the mouse pointer over it.

Procedure:

Preliminary adjustments:

1. Turn the telescope towards the white wall or screen and looking through the eye-piece, adjust its position until the cross wires are clearly seen.
2. Turn the telescope towards a window and focus the telescope on a long distant object.
3. Place the telescope parallel to the collimator.
4. Place the collimator directed towards the mercury vapour lamp. Switch on the lamp.
5. Focus collimator slit using collimator focusing adjustment.
6. Adjust the collimator slit width.
7. Place the prism table, note that the surface of the table is just below the level of the telescope and collimator.
8. Place a spirit level on the prism table. Adjust the base levelling screw until the bubble come at the centre of the spirit level.
9. Clamp the prism holder.
10. Clamp the prism in which the sharp edge is facing towards the collimator, and the base of the prism is at the clamp.

Least Count of Spectrometer

One main scale division (N) = ... minute
Number of divisions on vernier (V) = ...
L.C = $\frac{N}{V}$ = ... minutes

To determine the angle of the Prism:

1. The prism table is rotated in which the sharp edge of the prism is facing toward the collimator.
2. Rotate the telescope in one direction up to which the reflected ray is shown through the telescope.
3. Note corresponding main scale and vernier scale reading in both verniers (vernier I and vernier II).
4. Rotate the telescope in opposite direction to view the reflected image of the collimator from the second face of the prism.
5. Note the corresponding main scale and vernier scale reading in both vernier (vernier I and vernier II).
6. Find the difference between two readings, i.e., θ.
7. Angle of the prism, A= θ/2.

Reading of reflected ray from	Vernier 1			Vernier 2		
	MSR	VSR	Total	MSR	VSR	Total
Face 1 (say a)						
Face 2 (say b)						
Difference between a & b						

Mean θ = ... degrees
The angle of prism = ... degrees

To determine the Angle of minimum deviation:

1. Rotate the vernier table so that the light from the collimator falls on one face of the prism and emerges through the other face.
2. The telescope is turned to view the refracted image of the slit on the other face.
3. The vernier table is slowly turned in such a direction that the image of the slit moves directed toward the directed ray; i.e., in the direction of decreasing angle of deviation.
4. It will be found that at a certain position, the image is stationary for some moments. Vernier table is fixed at the position where the image remains stationary.
5. Note the readings on the main scale and vernier scale.
6. Carefully remove the prism from the prism table.
7. Turn the telescope parallel to the collimator, and note the direct ray readings.
8. Find the difference between the direct ray readings and deviated readings. This angle is called the angle of minimum deviation (D).
   The refractive index of the prism's material is determined by equation (3).

To determine the Dispersive power of prism:



1. Rotate the vernier table to have the light from the collimator fall on one face of the prism and emerged through another face. (refer to the given figure).
2. Turn the telescope to make the slit coincide with the telescope cross wire.
3. The emerged ray has different colours.
4. Slowly rotate the vernier table by using the vernier fine-adjusting slider.
5. Note the position where the coloured slits are stationary for some moment.
6. Turn the telescope to each colour, and note the readings for different colours.
7. Remove the prism, hence note direct ray reading.
8. Find the angle of minimum deviation for different colours. (Say, violet, blue, green, yellow).
9. Find the refractive index for these colours. Using equation (3).
10. Dispersive power for yellow and blue

$$\omega = \frac{(n_b - n_y)}{n - 1}$$

Where n_b and n_y are the refractive indices of blue and yellow, and $n = \frac{(n_b + n_y)}{2}$

Line	Vernier	Refracted Ray Readings	Direct readings	Difference (minimum deviation)	Mean D	n
Blue	V_1					
	V_2					
Green	V_1					
	V_2					
Yellow	V_1					
	V_2					
Red	V_1					
	V_2					

Dispersive power (ω) using blue and yellow line is

$$\omega = \frac{(n_b - n_y)}{n - 1}$$

Result:

Refractive index for blue = _____
Refractive index for green =_____
Refractive index for yellow = _____
Refractive index for red = _____
Dispersive power (ω) for _____ and _____ line = _____

Discussion:

Discuss on the possible errors in the experiment.